Understanding the Mechanisms of Fast Hyperparameter Transfer

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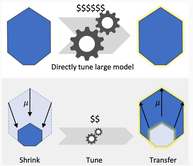
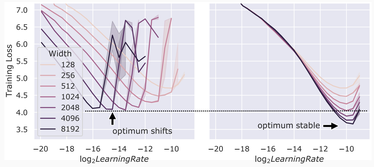
Introduction to μ-Transfer

μP lets you **tune a small proxy** and *transfer* HPs to a large run. *When does this work — and why?*



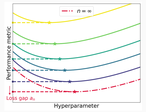
Shrink *to a small proxy,* ***tune*** *cheaply,* ***transfer*** *HPs to the big run.*



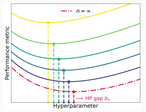
Standard parameterization (left): optimum shifts with width. μP (right): optimum stable.

Fast transfer

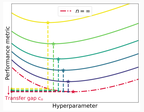
μP fixes optimal HPs in the width limit but says *nothing about rates*. Three gaps at width n:



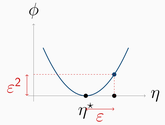
Loss gap a_n
finite vs. ∞ width



HP gap b_n
distance between best HPs



Transfer cost c_n
loss paid for wrong HP



Strongly-convex min
loss $\propto \varepsilon^2$ in HP error

Fast transfer: b_n shrinks *strictly faster* than $\sqrt{a_n}$ — beating the μP baseline $b_n = \mathcal{O}(\sqrt{a_n})$ set by strong convexity.

Useful transfer theorem

Direct: grid-search at the target width. **Transfer:** grid-search at small n, train once at target. Which wins?

Theorem. Suppose the gaps decay as power laws, $a_n \sim n^{-\alpha}$ and $b_n \sim n^{-\beta}$, near a strongly-convex minimum of the infinite-width loss. Then

transfer beats **direct** asymptotically $\iff$ $\beta > \frac{\alpha}{2}$, i.e., $b_n = o(\sqrt{a_n})$.

(same condition as *fast transfer*).

Why: transfer pays $c_n \propto b_n^2$ (HP error); direct pays $\sim a_n$ (loss gap). Transfer wins iff $b_n^2 = o(a_n)$.

Two synthetic testbeds: 2-layer ReLU learning a ball-indicator target (left); random-feature ridge regression in d dimensions (right).

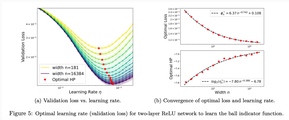
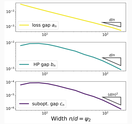


Figure 5: Optimal learning rate (validation loss) for two-layer ReLU network to learn the ball indicator function.

(a) Validation loss vs. learning rate. (b) Convergence of optimal loss and learning rate.

μP failure *— HP gap decays like $b_n \propto n^{-0.39} \approx \sqrt{a_n}$. Only the baseline; transfer breaks even.*



Provable fast transfer *— both $a_n, b_n \sim d/n$, so $b_n^2 = o(a_n)$. Theorem's condition holds.*

The mechanism: low-rank trajectory decomposition

Intuition: optimal HPs should depend on network statistics that stabilize with width *faster* than the loss itself. We decompose the stepwise linearized loss change via the eigenvalues of a gradient/update alignment matrix:

Matrix parameter $\boldsymbol{W} \in \mathbb{R}^{m\times n}$, gradient $\boldsymbol{G} \in \mathbb{R}^{m\times n}$, and update $\delta\boldsymbol{W} \in \mathbb{R}^{m\times n}$.

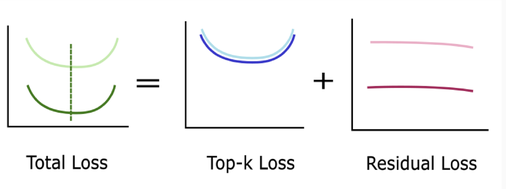
$$\boxed{\begin{aligned} \boldsymbol{A} &:= \tfrac{1}{2}\left(\boldsymbol{G}^\top \delta\boldsymbol{W} + \delta\boldsymbol{W}^\top \boldsymbol{G}\right) && \text{(alignment matrix)} \\ |\lambda_1| &\geq \cdots \geq |\lambda_n| && (\lambda_i \text{ eigenvalues of } \boldsymbol{A}) \end{aligned}}$$

Stepwise linearized loss change for parameter $\boldsymbol{W}$:

$$\delta\phi(\boldsymbol{W}) := \langle \boldsymbol{G}, \delta\boldsymbol{W}\rangle = \mathrm{Tr}(\boldsymbol{A}) = \underbrace{\sum_{i=1}^{k} \lambda_i}_{\delta\phi^k(\boldsymbol{W})} + \underbrace{\sum_{i=k+1}^{n} \lambda_i}_{\delta\phi^{-k}(\boldsymbol{W})}.$$

Summing over layers ℓ and time steps t:

$$\underbrace{\mathcal{L}(\boldsymbol{w}_T) - \mathcal{L}(\boldsymbol{w}_0)}_{\text{total loss}} \approx \underbrace{\sum_{\ell=1}^{L}\sum_{t=0}^{T-1} \delta\phi^k\left(\boldsymbol{W}_t^{(\ell)}\right)}_{\text{top-}k\text{ loss}} + \underbrace{\sum_{\ell=1}^{L}\sum_{t=0}^{T-1} \delta\phi^{-k}\left(\boldsymbol{W}_t^{(\ell)}\right)}_{\text{residual loss}}.$$



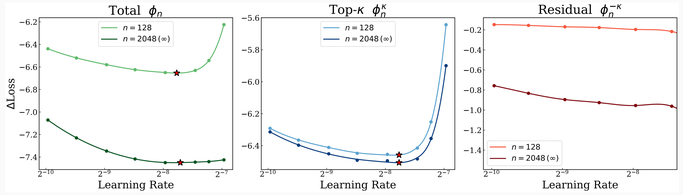
Cartoon: total loss = (U-shaped, width-stable) top-k + (flat) residual.

Conjecture. For some k: **(1)** the top-k loss is locally strongly convex at its optimum; **(2)** *top-k invariance* — the top-k loss and its optimal HP are approximately width-independent; **(3)** *residual flatness* — the residual loss is flat in HP, so it cannot move the optimum.

Experiments

We measure the decomposition on a real LLM training run, comparing the proxy width $n=128$ against the ∞-proxy $n=2048$ — the same pair as in the cartoon.

Setup: 4-layer LLaMA · WikiText-103 · Adam · warmup–stable–decay LR.



Total
The optimum *shifts* between the two widths — the μ-transfer baseline.

Top-k
Proxy top-k at $n = 128$ *overlaps* the infinite-width top-k and shares its optimum.

Residual
Flat in the learning rate at both widths — cannot move the optimum.

All three conjecture pieces (LSC, invariance, flatness) **verified** on a real LLM run.

In the paper: provable fast transfer for RF ridge regression; transfer behavior of Muon & Dion; a data-centric viewpoint of the top-k loss; a μP failure case. arXiv:2512.22768.

Takeaway. μP alone only guarantees HP transfer — not *useful* transfer. What makes tuning a small proxy worthwhile is a *low-rank structure in training*: a small, width-stable top-k component carries the HP signal, while the rest is flat.